

Lab 3: Architectural Pattern Selection — Written Justification

Problem Statement #49 — Webhook Ingestion & Retry Mechanism Hub

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We chose Microservices Architecture for the Webhook Ingestion & Retry Mechanism Hub.

Architectural Choice

The system is decomposed into five independently deployable services: the Webhook Ingestion Gateway, the Endpoint & Subscription Manager, the Delivery & Retry Engine, the Dead-Letter Queue Manager, and the Manual Recovery Console. Each owns a distinct responsibility from the requirements table (FR-001 through FR-005) and communicates with the others only through well-defined interfaces, which is the defining trait of a Microservices Architecture.

Two Scenario-Related Reasons

- Independent scaling under bursty load: NFR-001 requires the ingestion endpoint to sustain up to 2,500 requests/second, while the Delivery & Retry Engine is inherently slower — it deliberately waits through exponential backoff intervals (1s, 2s, 4s, 8s...) before re-attempting delivery (FR-002). Microservices let the stateless Ingestion Gateway be scaled horizontally during traffic spikes without over-provisioning the retry workers, and vice versa scale retry capacity during a downstream outage without touching ingestion throughput.
- Fault isolation for guaranteed durability: NFR-002 requires zero event loss even during a service restart. Because the Dead-Letter Queue Manager and Delivery & Retry Engine are separate services from the Ingestion Gateway, a backlog of failed deliveries or a crash in the retry/escalation path never blocks new inbound webhooks from being accepted and queued — ingestion keeps meeting its 99.999% reliability target (NFR-001) independently of the health of the delivery pipeline.

Security Advantage

Payload signature verification (FR-001) happens entirely inside the Webhook Ingestion Gateway before an event is ever enqueued. The Delivery & Retry Engine, Dead-Letter Queue Manager and Manual Recovery Console never receive raw, unauthenticated third-party traffic directly from the internet — only pre-validated events reach them through the IEventQueueInterface. This confines TLS termination, per-source rate limiting and signature checks to a single hardened boundary service, narrowing the attack surface for the rest of the pipeline.

Performance Benefit

Because the Ingestion Gateway and Delivery & Retry Engine are decoupled services communicating asynchronously through the IEventQueueInterface, the gateway can acknowledge and enqueue a webhook (HTTP 200 within 200ms, per FR-001's acceptance criteria) without waiting on downstream delivery latency or backoff timers. This decoupling of acceptance speed from delivery speed is what allows the gateway to sustain the 2,500 requests/second throughput target in NFR-001 even while individual deliveries are being retried in the background.